

Xi'an Jiaotong-Liverpool University

西交利物浦大學

PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT304		CPT	

2nd SEMESTER 2024/25 Final EXAMINATION

BACHELOR DEGREE – Year 4

Software Engineering II

TIME ALLOWED: 2 Hours + 10 min reading time

INSTRUCTIONS TO CANDIDATES

- 1、 This is a closed-book examination, which is to be written without books or notes.
- 2、 Total marks available are 100. This will count for 80% in the final marks.
- 3、 This exam consists of EIGHT (8) questions printed on 5 pages of papers including the cover page.
- 4、 Answer all questions. There is NO penalty for providing a wrong answer.
- 5、 Answers should be written in the answer booklet(s) provided.
- 6、 Only English solutions are accepted.
- 7、 All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Answer all questions. Write your answers in the provided answer booklet.

Question 1

Discuss how contemporary practices like Agile tools are used to tackle essential difficulties identified by Frederick Brooks. (20 marks)

Question 2

Below is the class design for a software application that manages a library system. This class violates the SOLID principles, specifically the Single Responsibility Principle (SRP) and the Open/Closed Principle (OCP).

```
1 public class LibraryManager {
2     private List<Book> books;
3
4     public LibraryManager()
5     {
6         this.books = new ArrayList<>();
7     }
8
9     public void addBook(Book book){
10        books.add(book);
11    }
12
13    public void removeBook(Book book){
14        books.removeBook(book);
15    }
16
17    public void printAllBooks(){
18        for(Book book: books){
19            // Code to prints book detail
20        }
21    }
22
23    public void saveBooksToFile(String filename){
24        // Code to save books to a file
25    }
26
27    public void loadBooksFromFile(String filename){
28        // Code to load books from a file
29    }
30 }
```

Modify your design to comply with the Open/Closed Principle (OCP), ensuring that the system can be extended with new features without modifying existing code. Your answer can be presented as either descriptive sentences or a class diagram. (8 marks)

Question 3

You are developing a navigation system for a mapping application. The system needs to support multiple routing algorithms, such as the shortest path, the fastest route, and a scenic route. Users should be able to switch between these routing strategies without modifying the existing system code at runtime.

Design a set of classes and interfaces to implement the Strategy pattern for the routing system. Include at least three different routing strategies: shortest path, fastest route, and scenic route. Your answer can be presented as either descriptive sentences or a class diagram. **(10 marks)**

Question 4

A software development team is designing a new online learning platform to support multiple user roles (students, instructors, administrators), manage course content, and integrate with external payment gateways and authentication services. The client requires the platform to be highly maintainable to allow easy updates, scalable to handle growing numbers of users and features, and delivered within a four-month timeline. **(10 marks)**

4.1 Name one software reuse technique suitable for this project. **(2 marks)**

4.2 State two reasons why the technique you chose in 4.1 is suitable for this project. **(8 marks)**

Question 5

Fault Tree Analysis (FTA) is a top-down, deductive method used to identify and analyze the causes of a system failure, known as the "top event." It employs a tree-like diagram with logic gates (e.g., AND, OR) to connect basic events (root causes) to the top event, allowing engineers to calculate the probability of failure and pinpoint critical risks. In this question, you will apply FTA to a software failure scenario, constructing a fault tree and computing probabilities. **(22 marks)**

Scenario

A real-time banking app experiences crashes during the online payment process, posing risks to the bank and its users while negatively affecting user trust. As a reliability engineer, you are tasked with analyzing the failure: "Payment processing crashes." Historical data and system design identify three independent causes—DB Failure, API Down, and Security Breach—any one of which can cause the payment process to crash. The causes and their probabilities are:

DB Failure (caused by both):

- "High DB load": 0.06/hour (6% chance per hour of overload).
- "No index": 0.25 (25% chance the database lacks indexes).

API Down (caused by either):

- "Network failure": 0.02/hour (2% chance per hour).
- "Server crash": 0.03/hour (3% chance per hour).

Security Breach (caused by both):

- "Hack attempt": 0.01/hour (1% chance per hour of an attack).
- "No firewall": 0.3 (30% chance no firewall is configured).

5.1 Build the fault tree. **(10 marks)**

5.2 Calculate the probability and MTTF for "Payment processing crashes". Show all intermediate steps (e.g., probabilities for "DB Failure," "API Down," "Security Breach") and the final probability, rounded to three decimal places. **(10 marks)**

5.3 Calculate the Mean Time To Failure (MTTF) in hours, and round to two decimal places. **(2 marks)**

Question 6

What common challenges do open-source projects face, and how can they be mitigated?
(4 marks)

Question 7

Explain the social and cultural aspects of open-source development. (4 marks)

Question 8

Consider a software application that calculates the monthly payments of a car loan. The application accepts three inputs: (22 marks)

- Cost of the car: The cost should be a positive integer between \$1,000 and \$200,000.
- Loan period: The period of the loan in years should be a positive integer between 1 and 10 years.
- Interest rate: The annual interest rate should be a positive decimal between 0.01 (1%) and 0.2 (20%).

8.1 Create Normal Boundary Value cases to test the functionality of this application. (5 marks)

8.2 Create Robust Boundary Value cases to test the functionality of this application. (6 marks)

8.3 Worst-case Boundary Value Cases:

- List all values for each input based on the boundary value analysis approach (minimum, minimum+1, nominal, maximum-1, maximum). (3 marks)
- Calculate the total number of test cases for this analysis. (2 marks)
- Produce 15 sample test cases using these values to test the functionality of this application. (6 marks)

The end of the paper